

1. `BadKey` overrides `equals()` but does not override `hashCode()`, violating the equals/hashCode contract.
2. `BadKey.equals(Object o)` always returns `true`, even for null or unrelated object types.
3. `objs[5]` accesses index 5 of an array with length 2, causing `ArrayIndexOutOfBoundsException`.
4. `n.length()` is called on a null reference, causing `NullPointerException`.